

```

y > ...
import pandas as pd

# Customer transaction data
data = {
    "Customer": ["Amit", "Priya", "Rahul", "Sneha", "Karan", "Amit", "Priya"],
    "Total_Spend": [5000, 12000, 3000, 15000, 8000, 7000, 5000],
    "Visit_Frequency": [10, 20, 5, 25, 12, 15, 8]
}

df = pd.DataFrame(data)

print("Original Data:")
print(df)

# Customer Segmentation
def categorize(spend):
    if spend >= 10000:
        return "Premium"
    elif spend >= 5000:
        return "Regular"
    else:
        return "Basic"

df["Category"] = df["Total_Spend"].apply(categorize)

print("\nCustomer Categories:")
print(df)

# Group By Category
summary = df.groupby("Category").agg({
    "Total_Spend": ["sum", "mean"],
    "Visit_Frequency": ["sum", "mean"]
})

print("\nCategory Wise Analysis:")
print(summary)

```

Original Data:

	Customer	Total_Spend	Visit_Frequency
0	Amit	5000	10
1	Priya	12000	20
2	Rahul	3000	5
3	Sneha	15000	25
4	Karan	8000	12
5	Amit	7000	15
6	Priya	5000	8

Customer Categories:

	Customer	Total_Spend	Visit_Frequency	Category
0	Amit	5000	10	Regular
1	Priya	12000	20	Premium
2	Rahul	3000	5	Basic
3	Sneha	15000	25	Premium
4	Karan	8000	12	Regular
5	Amit	7000	15	Regular
6	Priya	5000	8	Regular

Category Wise Analysis:

Category	Total_Spend		Visit_Frequency	
	sum	mean	sum	mean
Basic	3000	3000.0	5	5.00
Premium	27000	13500.0	45	22.50
Regular	25000	6250.0	45	11.25